

1. Snapshot Data

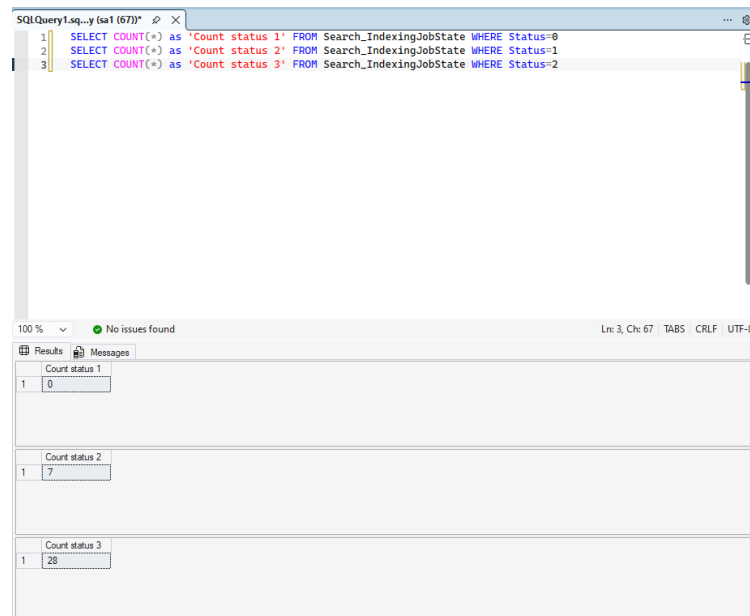


Figure 1. Snapshot of Search Indexing Job statuses before test execution

2. Failed Test Details

2.1 FR-01 Full build Elasticsearch

The Full Rebuild operation processed 6,902 records in total, including 4,000 Social Media records and 2,902 Platform Admin records. According to the test acceptance criteria, the resulting Elasticsearch index should contain 6,902 active documents.

A significant discrepancy was identified in the rebuilt index. While the Processed Count reached 6,902, Elasticsearch reported only 4,902 active documents (`docs . count`) and 2,000 deleted documents (`docs . deleted`).

The difference between the expected document count (6,902) and the actual active document count (4,902) is exactly 2,000 documents, which matches the number of deleted documents reported by Elasticsearch. This indicates that 2,000 documents were indexed but subsequently marked as deleted during or after the rebuild process.

As a result, the rebuilt index does not accurately reflect the total number of processed records, causing the validation criterion "Processed document count matches the total database records" to fail.

Expected Result: 6,902 processed records = 6,902 indexed documents.

Actual Result: 6,902 processed records $\neq$ 4,902 active indexed documents (`docs . count`) + 2,000 deleted documents (`docs . deleted`).

health	status	index	uuid	pri	rep	docs.count	docs.deleted	store.size	pri.store.size	dataset.size
green	open	.internal.alerts-transform.health.alerts-default-000001	ULzhPd2PTG2whIAMrEAK8A	1	0	0	0	250b	250b	250b
green	open	.internal.alerts-observability.logs.alerts-default-000001	n0EjwHVJRuG1MFQ2dWkC3TQ	1	0	0	0	250b	250b	250b
green	open	.internal.alerts-observability.uptime.alerts-default-000001	83-07KTJR4qDUmJrIiRCoA	1	0	0	0	250b	250b	250b
green	open	.internal.alerts-ml.anomaly-detection.alerts-default-000001	-xzcepaERx-12INwgUPZpA	1	0	0	0	250b	250b	250b
green	open	.internal.alerts-observability.slo.alerts-default-000001	Yh0-2EP05IakEnJHtE94YA	1	0	0	0	250b	250b	250b
green	open	.internal.alerts-observability.apm.alerts-default-000001	7r9uqIvRcFKAK7K7oTxw	1	0	0	0	250b	250b	250b
green	open	.internal.alerts-default.alerts-default-000001	aTi2jvjgSGW5u2BhqdrY5Q	1	0	0	0	250b	250b	250b
green	open	.internal.alerts-streams.alerts-default-000001	PvxcclJUSw-rmKS6PwG17Q	1	0	0	0	250b	250b	250b
green	open	.internal.alerts-security.attack.discovery.alerts-default-000001	H_YKYBYZTLKf1BtzqfA5aA	1	0	0	0	250b	250b	250b
green	open	.internal.alerts-observability.metrics.alerts-default-000001	-uizh_HgTNG9a9QSun9MZA	1	0	0	0	250b	250b	250b
green	open	.internal.alerts-ml.anomaly-detection-health.alerts-default-000001	dwzovQd4RLmVvQBLdbWYbQ	1	0	0	0	250b	250b	250b
green	open	.internal.alerts-observability.threshold.alerts-default-000001	Dr5b145BRqWd1qvBT5-Oog	1	0	0	0	250b	250b	250b
green	open	.internal.alerts-security.alerts-default-000001	XWqphU85QqCn5xv1vPt0g	1	0	0	0	250b	250b	250b
green	open	.internal.alerts-dataset.quality.alerts-default-000001	Ks76LHQgR9adFztaLQvSw	1	0	0	0	250b	250b	250b
yellow	open	rumble-content_v20260605040513	68FTBqKQ0S4d8ejcwZArQQ	1	1	4902	2002	3mb	3mb	3mb
yellow	open	rumble-content_v20260605040632	4r2-1SZu8Si4nnczkP8Cegw	1	1	4902	2000	3.2mb	3.2mb	3.2mb
green	open	.internal.alerts-stack.alerts-default-000001	dEOCCCHR5zGTj90x4wLQkw	1	0	0	0	250b	250b	250b

Figure 2. Comparison of the old and newly rebuilt Elasticsearch indices after full reindex execution.

2.2 FR-02 Alias Swap Validation

Following the Full Rebuild operation, the alias `rumble-content` was successfully redirected to the newly created index `rumble-content_v20260605040632`, indicating that the alias swap process was executed without interrupting search availability.

However, the validation criteria required not only a successful alias swap but also the removal of the legacy index after the swap completed. As shown in the index and alias mapping results, the previous index `rumble-content_v20260605040513` remained present in the Elasticsearch cluster even after the new index had been fully built and assigned to the alias.

Although all search requests were routed through the new index via the `rumble-content` alias, the legacy index was not purged as expected. This behavior violates the acceptance criterion stating that the old index should be removed after a successful swap.

Validation Result: Alias redirection completed successfully, but the legacy index remained in the cluster after rebuild completion. Therefore, the alias swap validation test was marked as Failed.

alias	index	filter	routing	index	routing	search
is_write_index		-	-	-	-	-
.alerts-transform.health.alerts-default	.internal.alerts-transform.health.alerts-default-000001	-	-	-	-	true
.alerts-ml.anomaly-detection.alerts-default	.internal.alerts-ml.anomaly-detection.alerts-default-000001	-	-	-	-	true
.alerts-observability.slo.alerts-default	.internal.alerts-observability.slo.alerts-default-000001	-	-	-	-	true
.alerts-streams.alerts-default	.internal.alerts-streams.alerts-default-000001	-	-	-	-	true
.alerts-observability.apm.alerts-default	.internal.alerts-observability.apm.alerts-default-000001	-	-	-	-	true
.alerts-default.alerts-default	.internal.alerts-default.alerts-default-000001	-	-	-	-	true
.kibana_security_solution	.kibana_security_solution_9.3.3_001	-	-	-	-	-
.kibana_security_solution_9.3.3	.kibana_security_solution_9.3.3_001	-	-	-	-	-
.alerts-security.attack.discovery.alerts-default	.internal.alerts-security.attack.discovery.alerts-default-000001	-	-	-	-	true
.alerts-observability.metrics.alerts-default	.internal.alerts-observability.metrics.alerts-default-000001	-	-	-	-	true
.alerts-ml.anomaly-detection-health.alerts-default	.internal.alerts-ml.anomaly-detection-health.alerts-default-000001	-	-	-	-	true
.alerts-security.alerts-default	.internal.alerts-security.alerts-default-000001	-	-	-	-	true
.siem-signals-default	.internal.alerts-security.alerts-default-000001	-	-	-	-	false
.alerts-stack.alerts-default	.internal.alerts-stack.alerts-default-000001	-	-	-	-	true
.kibana_analytics	.kibana_analytics_9.3.3_001	-	-	-	-	-
.kibana_analytics_9.3.3	.kibana_analytics_9.3.3_001	-	-	-	-	-
.kibana_usage_counters	.kibana_usage_counters_9.3.3_001	-	-	-	-	-
.kibana_usage_counters_9.3.3	.kibana_usage_counters_9.3.3_001	-	-	-	-	-
.alerts-observability.logs.alerts-default	.internal.alerts-observability.logs.alerts-default-000001	-	-	-	-	true
.kibana	.kibana_9.3.3_001	-	-	-	-	-
.kibana_9.3.3	.kibana_9.3.3_001	-	-	-	-	-
.alerts-observability.uptime.alerts-default	.internal.alerts-observability.uptime.alerts-default-000001	-	-	-	-	true
.kibana_alerting_cases	.kibana_alerting_cases_9.3.3_001	-	-	-	-	-
.kibana_alerting_cases_9.3.3	.kibana_alerting_cases_9.3.3_001	-	-	-	-	-
.security	.security-7	-	-	-	-	-
.kibana_search_solution	.kibana_search_solution_9.3.3_001	-	-	-	-	-
.kibana_search_solution_9.3.3	.kibana_search_solution_9.3.3_001	-	-	-	-	-
.alerts-observability.threshold.alerts-default	.internal.alerts-observability.threshold.alerts-default-000001	-	-	-	-	true
.kibana_task_manager	.kibana_task_manager_9.3.3_001	-	-	-	-	-
.kibana_task_manager_9.3.3	.kibana_task_manager_9.3.3_001	-	-	-	-	-
.alerts-dataset.quality.alerts-default	.internal.alerts-dataset.quality.alerts-default-000001	-	-	-	-	true
.inference-alias	.inference	-	-	-	-	true
rumble-content	rumble-content_v20260605040513	-	-	-	-	-
rumble-content	rumble-content_v20260605040632	-	-	-	-	-
.kibana_ingest	.kibana_ingest_9.3.3_001	-	-	-	-	-
.kibana_ingest_9.3.3	.kibana_ingest_9.3.3_001	-	-	-	-	-

Figure 3. Alias mapping after Full Rebuild execution

2.3 FR-03 Fetch feed data post-rebuild

Following the completion of the Full Rebuild process, the feed search API was executed to verify data availability and consistency in the newly rebuilt Elasticsearch index. The request was sent to the `/api/search/feed` endpoint using the predefined test parameters and sample identifiers.

The API returned **HTTP 200 OK**, indicating that the search service remained available and functional after the rebuild operation. The response payload contained the expected feed records, including the predefined sample content identifiers and associated metadata.

A comparison between the returned results and the reference dataset confirmed that the **top N query results matched the expected sample identifiers exactly**, with no missing, duplicated, or inconsistent records observed.

This result demonstrates that the rebuilt index is serving search requests correctly and that the Full Rebuild process did not negatively impact feed retrieval functionality.

Validation Result: HTTP 200 OK was returned, and the response payload matched the expected sample data. Therefore, the post-rebuild feed retrieval validation was marked as **Passed**.

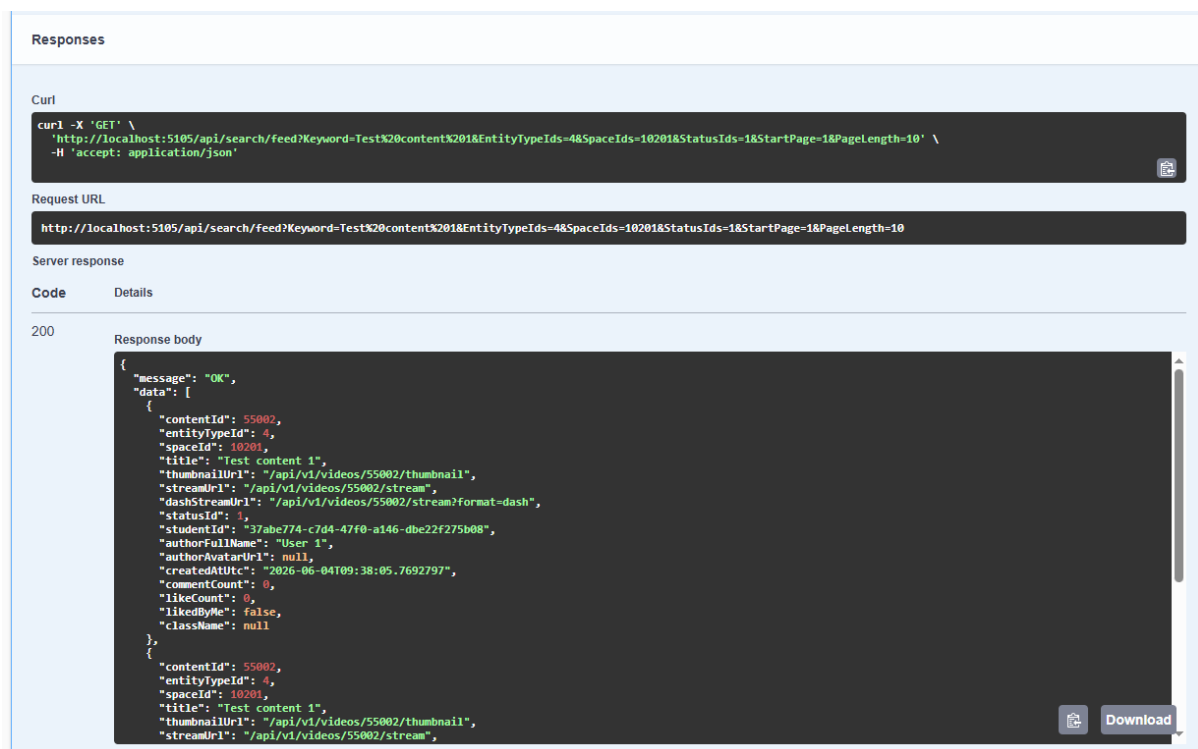


Figure 4. Feed data retrieval validation after Full Rebuild execution

2.4 FR-05 Reconcile record counts between Database and Elasticsearch.

Record count reconciliation identified a substantial mismatch between the number of records processed during the Full Rebuild operation and the number of active documents stored in Elasticsearch.

The rebuild job successfully processed **6,902 records**, while Elasticsearch reported only **4,902 active documents**. This resulted in a discrepancy of **2,000 records**,

equivalent to approximately **29.0%** of the processed count, which exceeds the acceptable threshold of **0.5%** defined by the test criteria.

Further investigation revealed that Elasticsearch also reported **2,000 deleted documents (docs.deleted)**, indicating that a significant number of documents were indexed and subsequently replaced during the rebuild process.

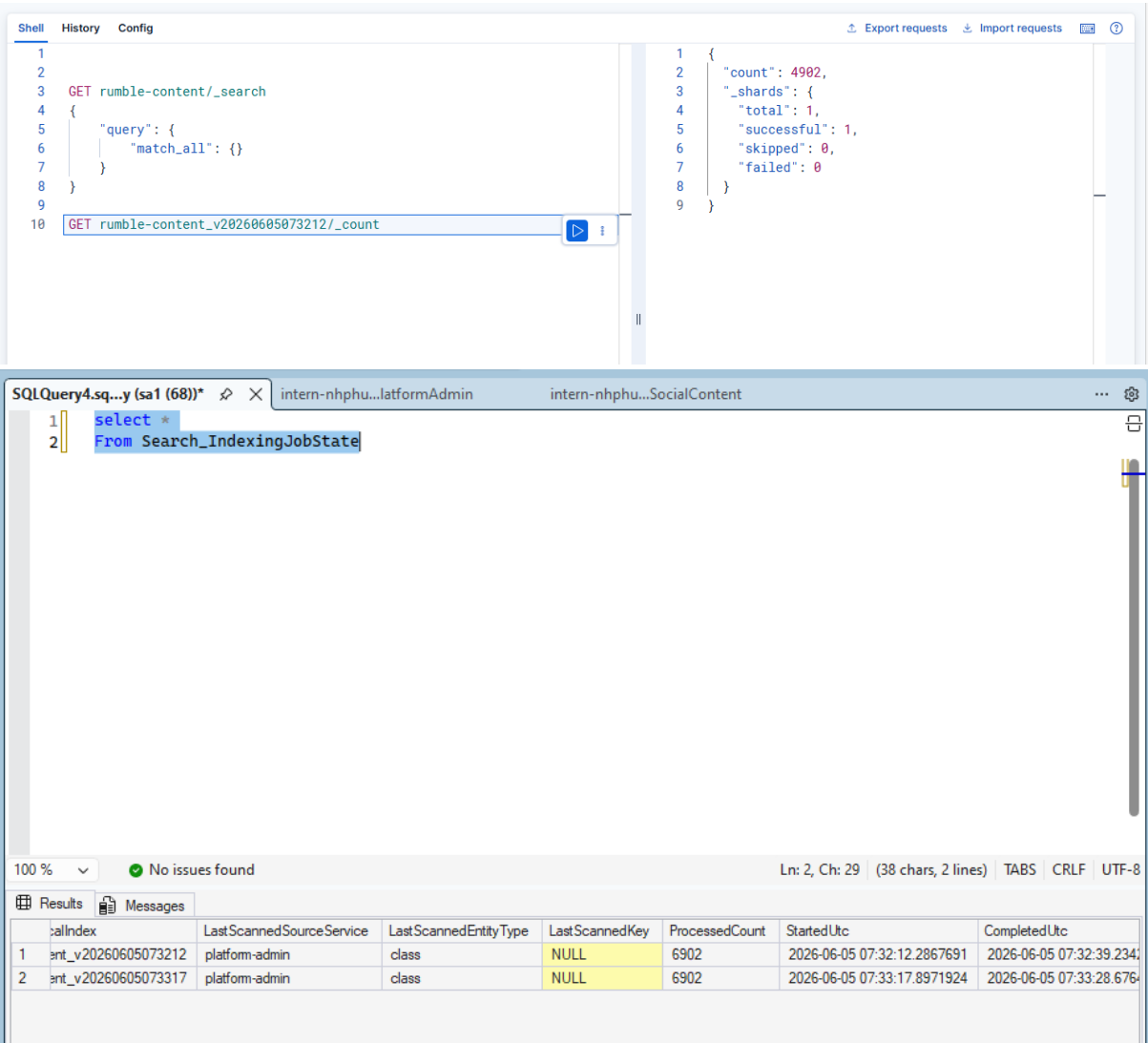


Figure 5. Database-to-Elasticsearch record count reconciliation showing significant discrepancy

Root cause analysis identified that the reindexing logic was executed multiple times for the same content entities. Specifically, both **Image** and **Video** indexing operations invoked the reindex process using the same Elasticsearch document identifier. When a document with an existing identifier was indexed again, Elasticsearch performed an update operation by creating a new version of the document and marking the previous version as deleted. As a result, the old documents were moved to the `docs.deleted` count while only the latest versions remained active in the index.

The source code configuration below shows that multiple entity types are included in the export targets, causing duplicate indexing operations for the same content records:

- Social Content → Image Entity
- Platform Admin → User Entity
- Platform Admin → Channel Entity
- Platform Admin → Class Entity

Consequently, although **6,902 records were processed successfully**, only **4,902 active documents remained searchable in Elasticsearch**, while **2,000 previous document versions were recorded as deleted**.

Expected Result: Database record count and Elasticsearch document count differ by no more than 0.5%.

Actual Result: Processed Count = 6,902, Active Elasticsearch Documents = 4,902, Deleted Documents = 2,000, resulting in an unacceptable discrepancy that caused the test case to fail.

Failure Reason: Duplicate reindex operations caused existing Elasticsearch documents to be overwritten. Elasticsearch retained the latest document versions and marked the previous versions as deleted, leading to a significant mismatch between processed records and active indexed documents.

```
private static readonly (string Source, string EntityType)[] ExportTargets =
[
    (SearchIndexConstants.SocialContentSource, SearchIndexConstants.RumbleContentEntity),
    (SearchIndexConstants.SocialContentSource, SearchIndexConstants.ImageEntity),
    (SearchIndexConstants.PlatformAdminSource, SearchIndexConstants.UserEntity),
    (SearchIndexConstants.PlatformAdminSource, SearchIndexConstants.SchoolEntity),
    (SearchIndexConstants.PlatformAdminSource, SearchIndexConstants.ClassEntity),
];
```

Figure 6. Reindex configuration contributing to duplicate document indexing

2.5 Perform continuous GET requests to api/search/feed during the rebuild process.

This test validated the system's ability to handle database CRUD operations and maintain data synchronization while a Full Rebuild operation was in progress.

During the rebuild process, database transactions were committed successfully without blocking or causing service interruption. Newly created and updated records were persisted in the database as expected.

After the rebuild completed, Elasticsearch automatically detected the pending changes and synchronized the affected records without requiring manual intervention. A synchronization delay of approximately **10–20 seconds** was observed between the database update and the appearance of the corresponding documents in Elasticsearch, which is consistent with the expected asynchronous indexing behavior.

The test confirmed that the rebuild process did not interfere with normal CRUD operations and that data consistency between the database and Elasticsearch was maintained through the automatic synchronization mechanism.

Expected Result: Database transactions commit successfully, synchronization status is marked as Pending, and automatic indexing is triggered after rebuild completion.

Actual Result: Database updates were committed successfully, and Elasticsearch automatically synchronized the new data after a short indexing delay.

Validation Result: CRUD operations and post-rebuild synchronization functioned as expected. The test case was marked as **Passed**.

SQLQuery4.sq...y (sa1 (68))* intern-nhphu...latformAdmin intern-nhphu...SocialContent

```

1 CREATE OR ALTER PROCEDURE usp_SeedRebuildTestData
2 AS
3 BEGIN
4     SET NOCOUNT ON;
5
6     -- Khai báo trước các biến ở đầu batch để tránh lỗi trùng lặp/scope
7     DECLARE @school_id int;
8     DECLARE @i int = 1;
9     DECLARE @uid nvarchar(450);
10    DECLARE @email nvarchar(256);
11    DECLARE @slug nvarchar(220);
12    DECLARE @class_id int;
13    DECLARE @student_id nvarchar(450);
14    DECLARE @vid_class_id int;
15    DECLARE @content_id int;
16
17    DECLARE @userIds TABLE (RowNum int IDENTITY(1,1), Id nvarchar(450));
18    DECLARE @classIds TABLE (RowNum int IDENTITY(1,1), Id int);
19
20    -- Lấy SchoolId đang tồn tại
21    SET @school_id = (SELECT TOP 1 Id FROM Rumble_Spaces WHERE ParentSpaceId = 0);
22
23    IF @school_id IS NULL
24    BEGIN
25        PRINT 'LỖI: Không tìm thấy School nào trong Rumble_Spaces. Vui lòng tạo ít nhất 1 school trước.';
26        RETURN;
27    END

```

100 % No issues found Ln: 82, Ch: 39 SPC CRLF UTF-8

Messages

Commands completed successfully.

Completion time: 2026-06-06T16:43:21.3541677+07:00

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☐ Name ↕	Health ↕	Status ↕	Primaries ↕	Replicas ↕	Documents ...	Storage ... ↕	Data stre... ↕
☐ rumble-content_v20260605094806	● yellow	open	1	1	4,922	3.22mb	

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Figure 7. Duplicate indices generated during the rebuild process

2.6 Possible Bug

During the investigation, a potential issue was identified in the index creation process.

Under the expected blue-green indexing strategy, the rebuild operation should create a new versioned index (for example, `rumble-content_v{timestamp}`) and perform all indexing activities against that versioned index before switching the alias.

However, the current implementation appears to create or index directly into the physical index `rumble-content` instead of exclusively using a versioned index. This behavior may bypass the intended blue-green workflow and increase the likelihood of duplicate indexing operations during rebuild and synchronization processes.

When the same document is indexed multiple times using the same document identifier, Elasticsearch replaces the existing document and records the previous version as deleted. This can lead to an increased `docs.deleted` count, document count discrepancies, and difficulties in validating rebuild results.

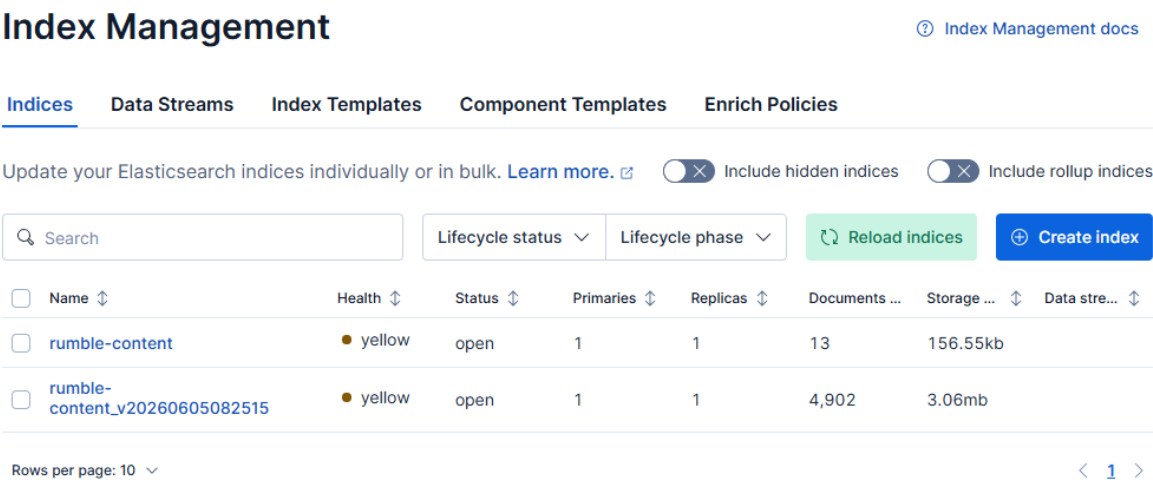


Figure 8. Index creation behavior observed during Full Rebuild